# 7.1 Continuous Probability Distribution

## 1. Continuous Random Variables

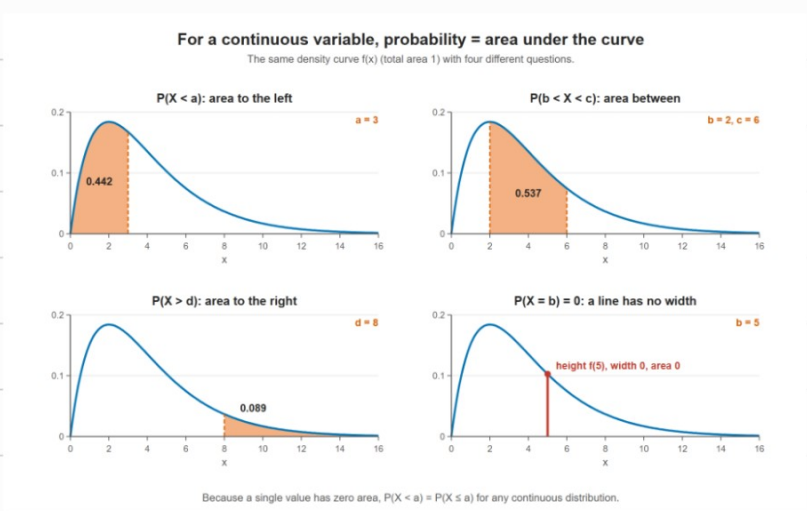
A **continuous** random variable $X$ can take **any** value in range, such as 2, 3.5 or 4.68. A **discrete** one is restricted to specific values, such as 1, 2, 3, ...

The curve of a continuous distribution is its **Probability Distribution Function (PDF)** $f(x)$. Its height above the x-axis at $x$ is $f(x)$.

## 2. The Key Rules

1. $f(x) \geq 0$ everywhere: The curve never dips below the x-axis
2. **Total area under the curve = 1.**
3. **Probability = Area under the curve** over the interval you want.
   - $P(X < a)$ = area to the **left** of $a$
   - $P(b < X < c)$ = area **between** $b$ and $c$
   - $P(X > d)$ = area to the **right** of $d$
4. **A Single Value has Zero Probability**: $P(X = b) = 0$. A vertical line has height but no width, so its area is 0.

The **normal distribution** (bell curve) is the most important example. It is covered in Chapter 8.




Every Probability Question about continuous variable is shaded area under the density curve.

A Single Value is a vertical line with no width, so its probability is zero 0.

## 3. Uniform Distribution

$f(x)$ is Constant between $a$ and $b$, so the graph is a rectangle with area 1.

Example: a uniform distribution from $a=2$ to $b=6$ has base $= 4$, so the hight must be $1/4$ for the area equal to $4 \times \frac{1}{4} = 1$

| PDF | Mean | Standard Deviation |
|---|---|---|
| $f(x) = \frac{1}{b-a}$ for $a \leq x \leq b$ | $\mu = \frac{a+b}{2}$ | $\sigma = \frac{b-a}{\sqrt{12}}$ |

## 4. Exponential Distribution

The Curve Starts at height $\lambda$ on the y-axis and decays as x increase

| PDF | Rate | Left Area | Right Area |
|---|---|---|---|
| $f(x) = \lambda e^{-\lambda x}$ | $\lambda = \frac{1}{\mu}$ | $P(X<x) = 1-e^{-\lambda x}$ | $P(X>x) = e^{-\lambda x}$ |

## 5. Key Takeaways

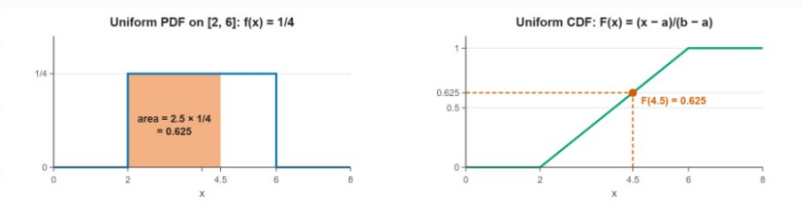
- Continuous Distribution are described by a PDF $f(x) \geq 0$ with a total area of 1.
- Probability = area under the curve
- $P(X = \text{one exact value}) = 0$, so "$<$" and "$\leq$" are Interchangeable.
- Uniform: A flat rectangle at height $\frac{1}{(b-a)}$. Exponential: a decaying curve with $\lambda = \frac{1}{\mu}$

# 7.2 Cumulative Distribution Functions and Probability Density Functions

## 1. PDF vs CDF

| | PDF, Probability Density Function | CDF, Cummulative Distribution Function |
|---|---|---|
| Notation | $f(x)$ | $F(x) = P(X \leq x)$ |
| Tells you | The Shape of distribution and the curve's height at x | The area to the left of x, the probability accumulates up to x |
| Range | $f(x) \geq 0$ | $0 \leq F(x) \leq 1$ |

## 2. Uniform Distribution on [a, b]



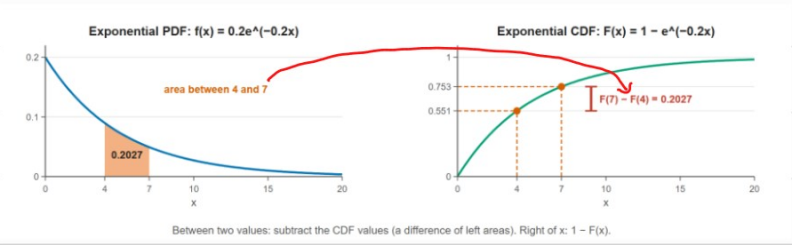

PDF: (a Constant height; no x in it)

$$f(x) = \frac{1}{b-a}$$

CDF: (the area of the rectangle from a to x = base x height):

$$F(x) = P(X \leq x) = (x-a) \cdot \frac{1}{(b-a)} = \frac{x-a}{b-a}$$

## 3. Exponential Distribution




* Picture: the region left of b minus the region left of a leaves the region between a and b.

PDF: (it starts at height $\lambda$ and decays, with $\lambda = \frac{1}{\mu}$):

$$f(x) = \lambda e^{-\lambda x}$$

CDF: (area to the left)

$$F(x) = P(X \leq x) = 1 - e^{-\lambda x}$$

Area to the right: (not a CDF, because the CDF is always the left area)

$$P(X \geq x) = 1 - F(x) = 1 - (1 - e^{-\lambda x}) = e^{-\lambda x}$$

Area between a and b: Substract two left areas.

$$P(a \leq X \leq b) = F(b) - F(a) = (1 - e^{-\lambda b}) - (1 - e^{-\lambda a}) = e^{-\lambda a} - e^{-\lambda b}$$

## 4. Reminders for Continuous Distributions

- $P(a \leq X \leq b) = P(a < X < b)$ because the endpoints don't change the area.
- $P(X = a) = 0$: A single line has height but no width

## 5. Key Takeaways

- PDF $f(x)$ gives the shape and height. CDF $F(x)$ gives the area to the left (Cumulative Probability).
- Right area: $1 - F(x)$. Area between $= F(b) - F(a)$
- Uniform $F(x) = \frac{x-a}{b-a}$. Exponential: $F(x) = 1 - e^{-\lambda x}$

# 7.3 Probability of a Uniform Distribution

## 1. Formulas (Uniform on $[a,b]$)

The graph is rectangle with total area 1:

area = base × height = $(b-a)f(x) = 1 \rightarrow f(x) = \frac{1}{b-a}$, $a \le x \le b$

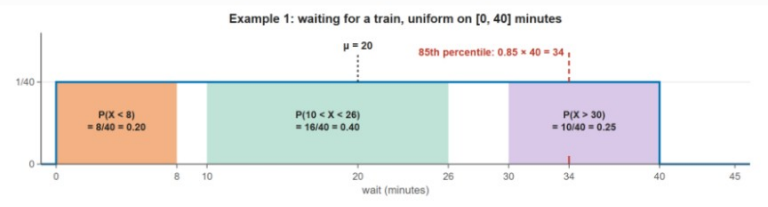
| Quantity | Formula |
|---|---|
| Probability of an Interval | Base × Height = $\frac{\text{Interval Width}}{b-a}$ |
| Mean = median | $\frac{a+b}{2}$ |
| Variance | $\frac{(b-a)^2}{12}$ |
| Standard Deviation | $\frac{b-a}{\sqrt{12}}$ |
| $k$-th percentile Value $K$ | $\frac{k}{100} = \frac{K-a}{b-a} \rightarrow K = a + \frac{k}{100}(b-a)$ |

Special Cases:

- $P(X=c)=0$ for any single value (a line has no width)
- $P(X>b) = P(X<a) = 0$ (the height is zero outside $[a,b]$).

## 2. Example 1 - Waiting for a train (0 to 40 minutes).

The wait time is Uniformly distributes on $[0.40]$.




### (a) PDF

$$f(x) = \frac{1}{b-a} = \frac{1}{40-0} = \frac{1}{40} \quad 0 \le x \le 40$$

(b) Graph: A flat line at height $\frac{1}{40}$ from $x=0$ to $x=40$, which forms a rectangle.

(c) Wait less than 8 minutes:

$$P(X<8) = 8 \times \frac{1}{40-0} = 0.20 = 20\%$$

(d) Wait more than 30 minutes:

$$P(X>30) = (40-30) \times \frac{1}{40} = \frac{10}{40} = 0.25 = 25\%$$

(e) Assorted Probabilities

- Wait between 10 and 26 minutes

  $P(10<X<26) = \frac{26-10}{40} = \frac{16}{40} = \frac{2}{5} = 40\%$.

- $P(X=20)=0$, a single value.
- $P(X>45)=0$, outside the range

(f) Mean and Standard Deviation

$\mu = \frac{0+40}{2} = 20$, $\sigma = \frac{40}{\sqrt{12}} = 11.55$

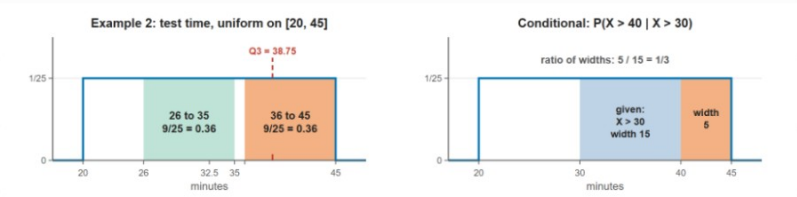
(g) 85th Percentile

The area to the left of K must be 0.85:

$0.85 = (K-0)\cdot\frac{1}{40} \rightarrow K = 0.85 \times 40 = 34$ Minutes

This is reasonable: The 75th Percentile is 30, so the 85th must lie between 30 and 40.

## 3. Example 2 - Time to Finish a Chemistry test (20 to 45 minutes)




(a) PDF

$f(x) = \frac{1}{45-20} = \frac{1}{25}$, $20 \le x \le 45$

(b) Graph: a rectangle of height $\frac{1}{25}$ from 20 to 45.

(c) More than 36 Minutes:

$P(X > 36) = \frac{45-36}{25} = \frac{9}{25} = 0.36 = 36\%$

(d) Between 26 and 35 Minutes:

$P(26 < X < 35) = \frac{35-26}{25} = \frac{9}{25} = 36\%$

The width is the same as in (c), so the probability is the same.

(f) Third Quartile ($Q_3$ = 75th Percentile)

$0.75 = (K-20) \times \frac{1}{25} \rightarrow K - 20 = 18.75 \rightarrow K = 38.75$ minutes.

Check: halfway between the median 32.5 and the maximum 45 is $(32.5+45)/2 = 38.75$

(g) Conditional: More than 40 minutes, given more than 30 minutes

$$P(X>40 \mid X>30) = \frac{P(X>40 \cap X>30)}{P(X>30)} = \frac{P(X>40)}{P(X>30)}$$

If $X>40$, then $X>30$ is automatically true, so the intersection is just $X>40$

$$P(X>40 \mid X>30) = \frac{(45-40)\times\frac{1}{25})}{(45-30)\times\frac{1}{25})} = \frac{5}{15} = \frac{1}{3} = 33.3\%$$

The height cancel, so conditional uniform probability is just a ratio of widths.

## 4. Key Takeaways

- Uniform Probability = width of the interval ÷ total width.
- A single point or anything outside $[a, b]$, has probability 0.
- Percentile: Solve $\frac{k}{100} = \frac{K-a}{b-a}$ for $K$
- Mean = Median = $\frac{(a+b)}{2}$ and $\sigma = \frac{(b-a)}{\sqrt{12}}$

# 7.4 Probability of an Exponential Distribution

## 1. Formulas

| Quantity | Formula |
|---|---|
| Rate Parameter | $\lambda = \frac{1}{\mu}$ |
| PDF | $f(x) = \lambda e^{-\lambda x}, \; x \geq 0$ |
| Area to the left | $P(X<x) = 1 - e^{-\lambda x}$ |
| Area to the right | $P(X>x) = e^{-\lambda x}$ |
| Area between a and b | $P(a<X<b) = P(X<b) - P(X<a) = e^{-\lambda a} - e^{-\lambda b}$ |

## 2. Example - Laptop LifeSpan

Laptops from Company XYZ last 5 years on average, and the lifespan follows an exponential distribution.

(a) Rate Parameter

$\lambda = \frac{1}{5} = 0.20$ per year

(b) PDF and graph

$f(x) = 0.20\, e^{-0.20x}$

The graph starts at the y-intercept $f(0) = 0.20 = \lambda$ (Since $e^0 = 1$) and decays towards 0 as x grows.

(c) Lasts less than 3 years (left area)

$P(X<3) = 1 - e^{-0.20(3)} = 1 - e^{-0.6} \approx 0.4512 = 45.12\%$

(d) Lasts more than 10 years (right area)

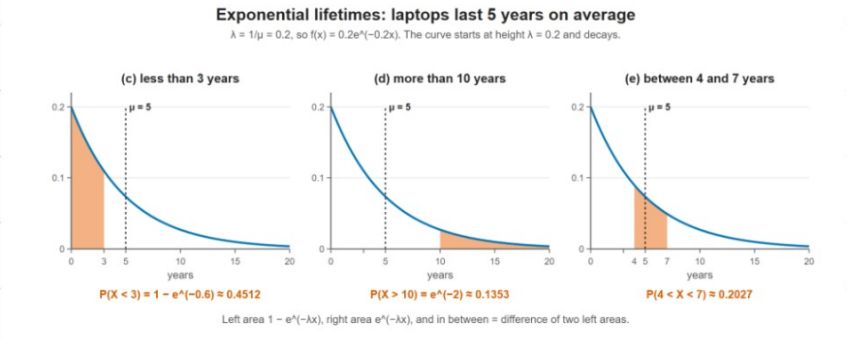
$P(X>10) = e^{-0.20(10)} = e^{-2} \approx 0.1353 = 13.53\%$

(e) Lasts between 4 and 7 years.

The area left of 7 minus the area left of 4 leaves the region between them.

$P(4<X<7) = (1 - e^{-1.4}) - (1 - e^{-0.8}) \approx 20.3\%$

(Equivalently, $e^{-0.8} - e^{-1.4} \approx 0.4493 - 0.2466 = 0.2027$)



## 3. Key Takeaways

- Find $\lambda = \frac{1}{\mu}$ first. The PDF's y-Intercept equal $\lambda$.
- Left area $1 - e^{-\lambda x}$; right area $e^{-\lambda x}$, between = difference of the left areas.
- The exponential distribution models waiting times and lifetimes. Short times are most likely, and long times become less and less likely.